

Wednesday Warm Up: Unit 0, Electrostatics I and II

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February 5, 2025

1 Memory Bank

- $\vec{F} = q\vec{E}$... Force on a charge q in the presence of an $\vec{E}$ -field.
- $\vec{F} = m\vec{a}$... Newton's 2nd Law.
- $\vec{E}_{\text{plane}} = \frac{\sigma}{2\epsilon_0} \hat{k} = \frac{Q}{2A\epsilon_0} \hat{k}$... The electric field of a plate or disk with charge Q and area A , for distances much smaller than the radius or length.
- **Mass of electron:** 9.11×10^{-31} kg.
- **Charge of electron:** 1.6×10^{-19} C.
- $\Delta U = q\Delta V$... Relationship between change in potential energy, charge, and voltage.
- $V(r) = kq/r$... Potential (voltage) of a point charge.
- $V(x) = Ex + V_0$... Linear relationship between voltage, electric field, and distance between two charge plates to form a capacitor.
- $(1/2)mv^2 = (3/2)kT$. Relationship between temperature and average atomic/molecular kinetic energy. T is the temperature in Kelvin, and $k = 1.38 \times 10^{-23}$ J/K, known as Boltzmann's constant.
- $Q = C\Delta V$... Relationship between stored charge, capacitance, and voltage.

2 Electric Fields

1. An evacuated tube uses an accelerating voltage of 40 kV to accelerate electrons to hit a copper plate and produce x rays. Non-relativistically, what would be the maximum speed of these electrons?
 2. Imagine a capacitor that has the following properties: the distance between the plates is 1 mm, the voltage in the middle is 0 Volts, $V_{\text{max}} = 6$ Volts, $V_{\text{min}} = -6$ Volts, and $\Delta V = 12$ V. (a) What is the electric field in between the plates? (b) If an electron is accelerated by the capacitor voltage, what is the kinetic energy in eV?
 3. Fusion probability is greatly enhanced when appropriate nuclei are brought close together, but mutual Coulomb repulsion must be overcome. This can be done using the kinetic energy of high-temperature gas ions or by accelerating the nuclei toward one another. (a) Calculate the potential energy of two singly charged nuclei separated by 1.0×10^{-12} m by finding the voltage of one at that distance and multiplying by the charge of the other. (b) At what temperature will atoms of a gas have an average kinetic energy equal to this needed electrical potential energy?
 4. Show that units of V/m and N/C for electric field strength are indeed equivalent.
 5. What charge is stored in a 180 μF capacitor when 120 V is applied to it?
2. Imagine a capacitor that has the following properties: the distance between the plates is 1 mm, the voltage